



SPARKBILLS · SPARKCURV TECHNOLOGIES

Sparkbills

Local Installation Guide — Windows 10 / 11

Step-by-step guide for a client (or on-site technician) to run Sparkbills on a single Windows PC without any cloud dependency. Uses Python 3.12, Node 20 LTS, MongoDB Community 7 and NSSM to auto-start on boot. Also covers thermal printer, barcode scanner, backups and the LAN-share option for multi-user setups.

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1. Requirements

- Windows 10 (build 19045+) or Windows 11.
- Minimum 8 GB RAM, 30 GB free disk. SSD recommended.
- Local administrator access.
- Modern browser (Chrome / Edge).
- Optional peripherals: 80mm thermal printer, USB barcode scanner, cash drawer with 24 V RJ11 trigger.

2. Choose your install path (single PC vs LAN)

Scenario	How to run	Access from
Single-user shop	Localhost only	This PC only (http://localhost:3000)
2 – 5 counters in same shop	One PC = server, others = browsers	LAN IP e.g. http://192.168.1.42:3000
Multiple branches over internet	Use the AWS EC2 guide instead	Cloud domain

3. Install Python 3.12

1. Open a browser → go to <https://www.python.org/downloads/windows/>.
2. Download the latest *Windows installer (64-bit)*.
3. Run the installer. On the first screen — **TICK** 'Add python.exe to PATH'.
4. Click 'Install Now'. Wait ~2 minutes.
5. Open a fresh PowerShell (Win+X → Windows Terminal) and verify:

```
python --version
# Python 3.12.x

pip --version
# pip 24.x from ...
```

4. Install Node.js 20 LTS + Yarn

1. Browse to <https://nodejs.org/en/download>.
2. Pick *LTS Windows Installer (64-bit)*.
3. Install with defaults (this also installs npm).
4. Open PowerShell as Administrator (right-click → 'Run as Administrator') and run:

```
node -v      # v20.x.x
npm -v       # 10.x.x
npm install -g yarn
yarn -v      # 1.22.x
```

5. Install MongoDB 7 Community + Compass

1. Go to <https://www.mongodb.com/try/download/community>. Pick MongoDB 7.0 — Windows — MSI.
2. Also download **MongoDB Compass** (GUI to browse your data).
3. Run the MongoDB installer → choose *Complete*.
4. On the 'Service Configuration' page — TICK 'Install MongoDB as a Service'. Data directory default (C:\Program Files\MongoDB\Server\7.0\data) is fine.
5. Finish the installer. MongoDB now auto-starts on every Windows boot.
6. Open Compass, click *Connect* with the default URI `mongodb://localhost:27017` — you should see 3 default databases (*admin*, *config*, *local*).

Note: Local install does NOT need MongoDB authentication — the DB is only reachable from the same PC.

6. Install Git for Windows

1. Browse to <https://git-scm.com/download/win> and install with default options.
2. Verify:

```
git --version    # git version 2.x.x.windows.x
```

7. Download the Sparkbills source

1. Choose an install folder, e.g. C:\Sparkbills.
2. Option A — from Git repo (open PowerShell in C:\):

```
cd C:\
git clone https://&lt;YOUR-REPO-URL&gt;.git Sparkbills
cd Sparkbills
```

- Option B — the vendor gave you a ZIP: right-click → Properties → tick *Unblock* → Extract-All to C:\Sparkbills.

8. Backend setup — venv, dependencies, .env

```
cd C:\Sparkbills\backend
python -m venv venv
.\venv\Scripts\Activate.ps1
pip install --upgrade pip
pip install -r requirements.txt
pip install emergentintegrations --extra-index-url `
  https://d33sy5i8bnduwe.cloudfront.net/simple/
```

If PowerShell refuses to run Activate.ps1:

```
# One-time (as Administrator):
Set-ExecutionPolicy -Scope CurrentUser RemoteSigned
```

Create C:\Sparkbills\backend\.env in Notepad with these exact contents:

```
MONGO_URL="mongodb://localhost:27017"
DB_NAME="sparkbills_local"
CORS_ORIGINS="http://localhost:3000"
JWT_SECRET="local-dev-secret-change-me-please"

ADMIN_EMAIL="admin@sparkcurv.in"
ADMIN_PASSWORD="Admin@2026"
ADMIN_NAME="SparkCurv Admin"
ADMIN_BUSINESS_NAME="SparkCurv Technologies"

EMERGENT_EMAIL_KEY="" # leave blank to disable email
EMAIL_FROM_NAME="Sparkbills"
```

Important: Save the file WITHOUT a .txt extension. If Notepad adds one, rename to just .env in Windows Explorer.

Test the backend once, interactively:

```
cd C:\Sparkbills\backend
.\venv\Scripts\Activate.ps1
uvicorn server:app --host 127.0.0.1 --port 8001

# Expected: 'Uvicorn running on http://127.0.0.1:8001'
# In a browser, visit http://localhost:8001/api/
# You should see {"app":"Sparkbills","vendor":"SparkCurv Technologies","status":"ok"}
```

Press Ctrl+C when done — you'll later run it as a service.

9. Frontend setup — build and serve

Create C:\Sparkbills\frontend\.env:

```
REACT_APP_BACKEND_URL="http://localhost:8001"
```

Install dependencies and produce a production build:

```
cd C:\Sparkbills\frontend
yarn install

# --- Option A (fastest for a single user): dev server ---
yarn start
# Browser auto-opens http://localhost:3000

# --- Option B (production build, faster runtime) ---
yarn build
npm install -g serve
serve -s build -l 3000
# Serves http://localhost:3000
```

10. First-run smoke test

1. With both backend and frontend running, open **http://localhost:3000** in Chrome / Edge.
2. The animated login page loads. Enter **admin@sparkcurv.in / Admin@2026**.
3. You land on the Dashboard.
4. Go to **Settings** → **Business** and fill in your GSTIN, address, UPI ID.
5. Go to **Settings** → **Users**, click your row → *Change Password*.
6. Create a test invoice from **Sales** → **New Invoice** to confirm PDF generation.

11. Auto-start on Windows boot with NSSM

NSSM ('Non-Sucking Service Manager') wraps any command as a Windows service so it starts on every boot.

1. Download NSSM from **<https://nssm.cc/download>** and extract nssm.exe to C:\Sparkbills\bin\.
2. Open PowerShell as Administrator, then create two services (backend + frontend):

```
cd C:\Sparkbills\bin

# Backend service
.\nssm.exe install SparkbillsBackend `
  "C:\Sparkbills\backend\venv\Scripts\python.exe" `
  "-m" "uvicorn" "server:app" "--host" "127.0.0.1" "--port" "8001"
.\nssm.exe set SparkbillsBackend AppDirectory "C:\Sparkbills\backend"
.\nssm.exe set SparkbillsBackend AppEnvironmentExtra `
  "PATH=C:\Sparkbills\backend\venv\Scripts;%PATH%"
.\nssm.exe start SparkbillsBackend

# Frontend service (serves the built React app)
.\nssm.exe install SparkbillsFrontend `
  "C:\Users\<YOU>\AppData\Roaming\npm\serve.cmd" `
  "-s" "C:\Sparkbills\frontend\build" "-l" "3000"
.\nssm.exe set SparkbillsFrontend AppDirectory "C:\Sparkbills\frontend"
.\nssm.exe start SparkbillsFrontend
```

Both services now auto-start on every Windows reboot. Manage them manually with:

- **services.msc** → find 'Sparkbills' entries → Start / Stop / Restart.
-

12. Windows Defender / Firewall rules

- First time uvicorn or node runs, Windows Firewall may prompt — click **Allow access** for *Private networks*.
- If other computers in the same LAN must access the app, allow it on *Public networks* too and use the PC's LAN IP (see Step 15).
- Windows Defender may quarantine nssm.exe — add C:\Sparkbills\bin to the exclusion list.

13. Thermal printer setup (80mm)

1. Plug the printer via USB or connect to Wi-Fi.
2. Install the manufacturer's Windows driver (typically named EPSON / TVSE / TP-806).
3. Set the printer as the default in **Windows Settings** → **Bluetooth & devices** → **Printers & scanners**.
4. In Chrome, when you open a Thermal PDF from Sparkbills, hit Ctrl+P and choose the printer. Paper size 80mm × 297mm.
5. For unattended printing (kitchen KOT), enable Chrome's 'Print without preview' kiosk mode.

14. Barcode scanner setup

Any USB barcode scanner that emulates a keyboard (HID mode) works — Honeywell, Zebra, TVS-E BSC-101, etc.

1. Plug the scanner in. Windows auto-detects it as a keyboard.
2. Open Notepad and scan any barcode — if the number appears, the device is ready.
3. In Sparkbills, open **Retail POS**. The barcode input field is focused by default. Scan → item added.
4. Configure the scanner (via its manual's config barcodes) to auto-append **Enter** after each scan.

15. LAN sharing to other computers

1. On the server PC, open **Command Prompt** and run `ipconfig`. Note the IPv4 address (e.g. 192.168.1.42).
2. Edit `C:\Sparkbills\frontend\.env` — change `REACT_APP_BACKEND_URL` to `http://192.168.1.42:8001`.
3. Edit `C:\Sparkbills\backend\.env` — change `CORS_ORIGINS` to `http://192.168.1.42:3000`.
4. In NSSM, edit `SparkbillsBackend` — change `--host 127.0.0.1` to `--host 0.0.0.0`.
5. Rebuild the frontend: `cd C:\Sparkbills\frontend; yarn build`.
6. Restart both services.
7. On the other PCs, open Chrome and visit **`http://192.168.1.42:3000`**.

Important: Give the server PC a static IP in your router so it doesn't change on reboot.

16. Daily local backup

```
# Create C:\Sparkbills\backup.ps1
$stamp = Get-Date -Format "yyyyMMdd-HH:mm"
$out = "C:\Sparkbills\backups\sparkbills-$stamp"
New-Item -ItemType Directory -Force -Path $out | Out-Null
& "C:\Program Files\MongoDB\Server\7.0\bin\mongodump.exe" `
  --db sparkbills_local --out $out
# Optional: copy to external drive
Copy-Item -Recurse $out "E:\Backups\Sparkbills\"
```

Schedule daily 22:00 via Task Scheduler (one-time command):

```
schtasks /Create /SC DAILY /ST 22:00 /TN SparkbillsBackup `
  /TR "powershell -ExecutionPolicy Bypass -File C:\Sparkbills\backup.ps1"
```


17. Troubleshooting

Symptom	Cause	Fix
PowerShell blocks Activate.ps1	Execution policy Restricted	Set-ExecutionPolicy -Scope CurrentUser RemoteSigned (as Admin).
Port 8001 or 3000 in use	Another program already running there	netstat -ano findstr :8001 → taskkill /PID <pid> /F
MongoDB service not running	Stopped	services.msc → find MongoDB Server → Start
Login redirects back to /login	CORS_ORIGINS mismatch (typo, missing scheme)	Fix backend/.env and restart the SparkbillsBackend service
Emails not sent	EMERGENT_EMAIL_KEY empty	Leave blank to disable email — other flows still work
Cannot log in	Credentials changed after seeding	Sign in with the ones you set last, or delete DB and re-seed
NSSM install fails	Not running as Administrator	Right-click PowerShell → Run as Administrator
'serve' command not found	Global npm bin not in PATH	Reopen PowerShell after npm install -g serve, or use full path

18. Uninstall

```
# Remove services
cd C:\Sparkbills\bin
.\nssm.exe remove SparkbillsBackend confirm
.\nssm.exe remove SparkbillsFrontend confirm

# Delete files
Remove-Item -Recurse -Force C:\Sparkbills

# Uninstall Python / Node / Mongo from 'Add or remove programs' (Settings &rarr; Apps).
```

For custom features, licensing or on-site setup support contact admin@sparkcurv.in.